

Integral

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$$I = \int_0^\infty W\left(\frac{1}{x^2}\right) dx$$

Find I , with $W(x)$ being the W Lambert function, defined as the inverse function of xe^x :

$$W(xe^x) = x$$

Let's substitute:

$$ue^u = \frac{1}{x^2} \tag{1}$$

$$x = e^{-\frac{1}{2}u} \frac{1}{\sqrt{u}} \tag{2}$$

$$dx = \frac{d}{du} \left(e^{-\frac{1}{2}u} \frac{1}{\sqrt{u}} \right) du \tag{3}$$

$$= \left(-\frac{1}{2\sqrt{u^3}} e^{-\frac{1}{2}u} - \frac{1}{2\sqrt{u}} e^{-\frac{1}{2}u} \right) du \tag{4}$$

$$= -\frac{1}{2} e^{-\frac{1}{2}u} \left(\frac{1}{\sqrt{u^3}} + \frac{1}{\sqrt{u}} \right) du \tag{5}$$

$$= -\frac{\sqrt{u}(1+u)}{2u^2} e^{-\frac{1}{2}u} du \tag{6}$$

Before substituting, let's check the boundaries:

$$\lim_{u \rightarrow 0} e^{-\frac{1}{2}u} \frac{1}{\sqrt{u}} = [\infty] \tag{7}$$

$$\lim_{u \rightarrow \infty} e^{-\frac{1}{2}u} \frac{1}{\sqrt{u}} = \sqrt{\frac{1}{ue^u}} = \frac{1}{[\infty]} = 0 \tag{8}$$

So, if x goes to 0, u must go to infinity, and if x goes to infinity, u must go to 0, so we will have to switch the boundaries of the integral when substituting. But first, for convenience, we shall write:

$$\alpha(u) = e^{-\frac{1}{2}u} \frac{1}{\sqrt{u}} \quad (9)$$

$$\frac{d}{du}\alpha(u) = -\frac{\sqrt{u}(1+u)}{2u^2}e^{-\frac{1}{2}u} \quad (10)$$

Now we can rewrite the integral:

$$I = \int_{\infty}^0 -W(ue^u) \frac{\sqrt{u}(1+u)}{2u^2} e^{-\frac{1}{2}u} du \quad (11)$$

$$= - \int_{\infty}^0 u \frac{\sqrt{u}(1+u)}{2u^2} e^{-\frac{1}{2}u} du \quad (12)$$

$$= \int_0^{\infty} u \left(-\frac{d}{du}\alpha(u) \right) du \quad (13)$$

Now let's integrate by parts:

$$I = -u\alpha(u) \Big|_0^{\infty} + \int_0^{\infty} \alpha(u) du \quad (14)$$

This leaves us with another integral to calculate:

$$\tilde{I} = \int_0^{\infty} \alpha(u) du = \int_0^{\infty} e^{-\frac{1}{2}u} \frac{1}{\sqrt{u}} du \quad (15)$$

That square root at the denominator resembles the derivative of the square root function itself, so let's substitute and hope for the best:

$$v = \sqrt{u} \quad \rightarrow \quad u = v^2 \quad (16)$$

$$du = (2v)dv \quad (17)$$

$$(18)$$

And then:

$$\tilde{I} = \int_0^{\infty} e^{-\frac{1}{2}v^2} \frac{1}{v} (2v) dv \quad (19)$$

$$= 2 \int_0^{\infty} e^{-\frac{1}{2}v^2} dv \quad (20)$$

Since the integrand is an even function, we can rewrite the integral with new boundaries:

$$\tilde{I} = \int_{-\infty}^{+\infty} e^{-\frac{1}{2}v^2} dv \quad (21)$$

And now, with one final substitution, the Gaussian integral will be in our hands!

$$w = \frac{v}{\sqrt{2}} \rightarrow v = \sqrt{2}w \quad (22)$$

$$dv = (\sqrt{2})dw \quad (23)$$

$$\tilde{I} = \sqrt{2} \int_{-\infty}^{+\infty} e^{-w^2} dw \quad (24)$$

$$= \sqrt{2\pi} \quad (25)$$

Now, let's head back to (14):

$$I = -\frac{u}{\sqrt{u}} e^{-\frac{1}{2}u} \Big|_0^{\infty} + \tilde{I} \quad (26)$$

$$= -\sqrt{\frac{u^2}{ue^u}} \Big|_0^{\infty} + \sqrt{2\pi} \quad (27)$$

$$= -\lim_{u \rightarrow \infty} \sqrt{\frac{u}{e^u}} + \lim_{u \rightarrow 0} \sqrt{\frac{u}{e^u}} + \sqrt{2\pi} \quad (28)$$

We can easily see that both limits are equal to 0, so this leaves us with our final answer:

$$I = \int_0^{\infty} W\left(\frac{1}{x^2}\right) dx = \sqrt{2\pi}$$

The integral was taken from <https://dailyintegral.com> and is the hard problem of 28 February 2026.